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Agentic AI Communication Assistant for Global Teams

Language barriers remain a major challenge in today's globalized world. Professionals work across countries, some students study in non-native languages, and immigrant families don't really understand important documents in unfamiliar languages. These situations often lead to misunderstandings and reliance on others for help. While translation tools exist, they usually only provide direct translations and not help users understand context, summarize long content, simplify complex wording, or read text from images and scanned documents. This project proposes an Agentic AI Communication Assistant for Global Teams, designed to offer practical and accessible language support through AI agents.

The real business problem addressed by this project is ineffective communication across language barriers in workplaces and educational settings. Miscommunication can reduce productivity, cause errors, and limit participation from individuals who are less confident in a second language. Outside of work, immigrant families may struggle to understand school or healthcare documents. The target users include global professionals, college students, and immigrant parents. These users need more than basic translation, but a need to understand information in a clear and adaptable language.

Communication support requires multiple steps, making an agentic AI approach suitable. The system must recognize the input type, extract text if needed, translate, and decide whether summarization, simplification, or audio output would help. For instance, a scanned document requires an optical character recognition (OCR) tool before translation, and long reports benefit

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from summaries. Coordinated agents allow the system to act like a communication assistant rather than a simple translator.

In use case scenario one, a professional receives a long document in another language. The system detects the language, extracts text, translates it, summarizes key points, and an optional speech module could read translation out loud. This helps the user quickly understand important information without needing a human translator.

In scenario two, an immigrant parent sees a notice. Using a phone camera, the parent captures an image of the notice. The system extracts the text using OCR, translates the notice and simplifies wording and provides an optional audio version for the parent to use. This supports real time, independent understanding.

Lastly, in use case scenario three, a professional receives an email in a second language. The system translates and summarizes the message, then helps draft a reply in the original language with an appropriate professional tone and context appropriate. This helps users communicate more effectively in global workplaces and reduce the risk of misrepresentation.

The initial system design includes several multiple agents coordinated by a Planner Agent. An Input Processing Agent identifies the input type, while an OCR/Document Agent extracts text when needed. A Language Detection Agent determines the source language, and a Translation Agent converts the text to the users preferred language. A Summarizer Agent that condenses long text, a Simplification Agent makes language easier to understand, and a Speech Agent provides audio output. These agents use tools as OCR libraries, translation models, and text-to-speech services.

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Although the project is feasible within the course timeframe, it presents several risks and challenges. Some of the challenges include translation inaccuracies, OCR errors, and limits when processing long documents. Maintaining tone across languages and avoiding sensitive data processing are also concerns. These risks can be reduced by breaking large texts into sections, allowing user review, and including clear disclaimers.

The project will be completed in structured phases across the remaining weeks of the course. The first phase finalizing system design, defining agent roles and selecting tools such as translation, OCR and text to speech. Next, I will build core translation features with language detection, followed by adding OCR support image and document inputs. Once basic translation works for multiple types of inputs, I plan to integrate a summarization agent and then a simplification agent if time is willing to make content easier to understand. Later, I will implement a planner agent to coordinate how different agents and tools are used. The final weeks will focus on system integration, testing and preparing demo and presentation, ensuring the project remains realistic and achievable within the course timeline.